

Nomenclature of heterocycles

I. seminar

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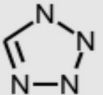
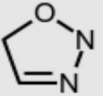
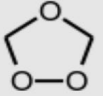
kornetmaryna@ukr.net

19.04.2023

Online Resources

<https://goldbook.iupac.org/indexes/rings>

The screenshot shows the 'IUPAC Gold Book' website. On the left is an 'Alphabetical Index' with a grid of letters A-Z and XYZ. Below it are 'Additional Indexes' including Physical Constants, Units of Measure, Physical Quantities, SI Prefixes, and Ring Index. The main section is the 'Ring Index', which lists three chemical structures with their formulas and the number of molecules:

Chemical Structure	Formula	Count
	CH_4N_4	1 molecule
	$\text{C}_2\text{H}_4\text{N}_2\text{O}$	2 molecules
	$\text{C}_2\text{H}_4\text{O}_3$	1 molecule

<https://www.degruyter.com/database/iupac/html>

The screenshot shows the 'IUPAC Standards Online' search page on the Degruyter website. It features a search bar with the text 'Search IUPAC Standards Online' and a 'What does this mean?' link. Below the search bar is a banner image with the text 'IUPAC STANDARDS ONLINE' and 'DE GRUYTER DATABASE'.

<https://www.acdlabs.com/resources/iupac-blue-book/>

The screenshot shows the 'IUPAC Nomenclature of Organic Chemistry' page on the ACD/Labs website. The page has a dark blue header with the ACD/Labs logo and navigation links: Products, Solutions, Analytical Chemistry Informatics, Technical Support, Resources, and Events & Seminars. The main content area is white with the title 'IUPAC Nomenclature of Organic Chemistry' in large, bold letters.

2.11 - The following trivial and semi-trivial names constitute a partial list of such names which are retained for the compound and as a basis of fusion names. The names of the radicals shown are formed according to Rule B-5.

https://www.acdlabs.com/iupac/nomenclature/79/r79_958.htm

2.12 - The following trivial and semi-trivial names are retained but are not recommended for use in fusion names. The names of the radicals shown are formed according to Rule B-5.

https://www.acdlabs.com/iupac/nomenclature/79/r79_959.htm

Introduction

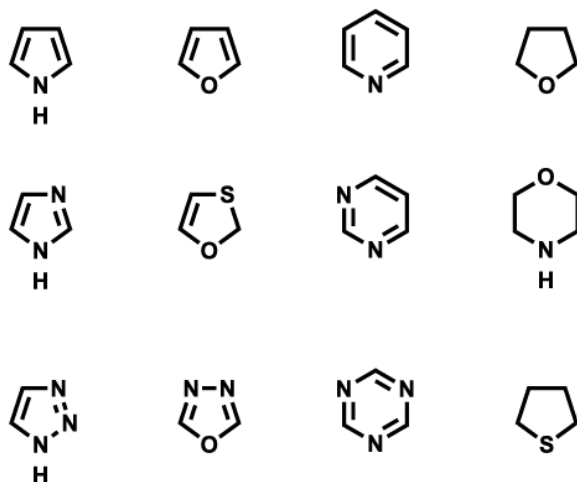
What is a Heterocycle?

“cyclic compounds having as ring members atoms of at least two different elements”



hetero → at least two different elements
cycle → cyclic compound

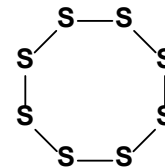
Some heterocycles: ✓



Not heterocycles: ✗



- Benzene is cyclic but all the atoms are carbon

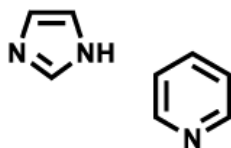


- S₈ is cyclic but all the atoms are sulfur

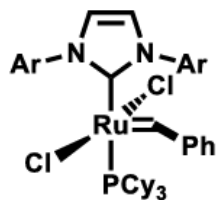
Importance of Heterocycles:

- Chemistry**

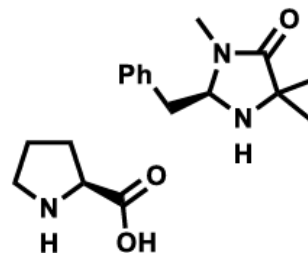
Heterocycles are key components of reagents, catalysts and chemical tools



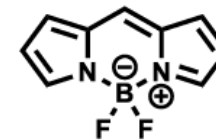
nucleophilic catalyst
and base



ligand
(Grubbs II metathesis)



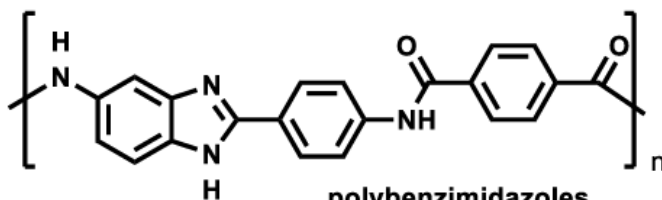
organocatalysts



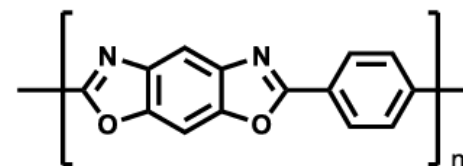
molecular imaging
(bodipy)

- Polymers**

Heterocycles are the key component in many important polymers



polybenzimidazoles
(flame retardant material)

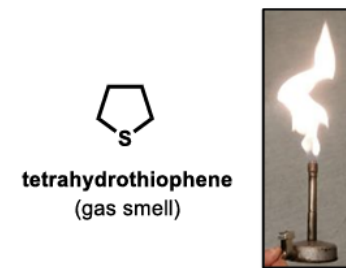
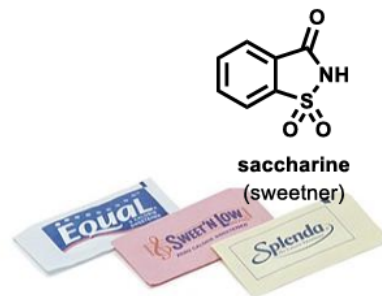
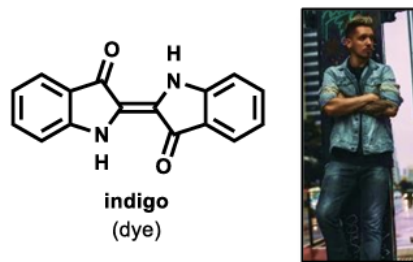


Zylon
(very strong fibres)



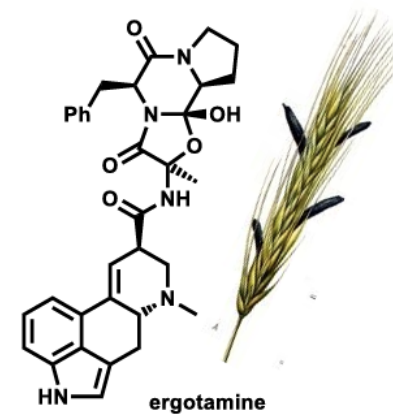
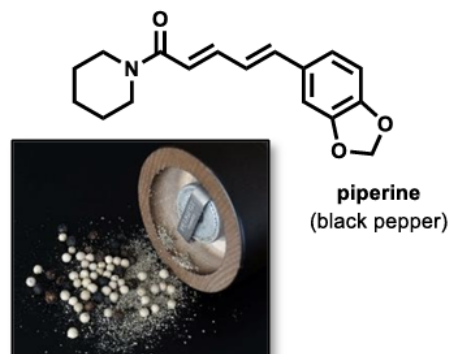
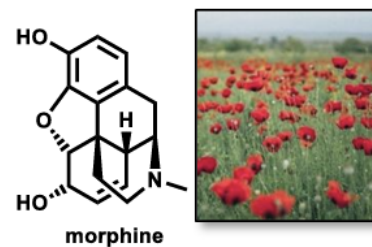
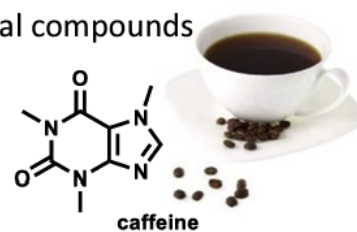
- **Everyday Items**

Heterocycles are the key component in everyday objects



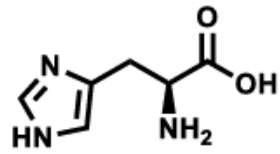
- **Natural Compounds**

Heterocycles are the key component in natural compounds

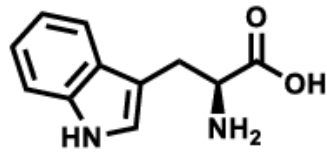


- **Nature's Building Blocks**

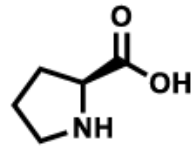
Heterocycles are found in 3 of the 20 proteinogenic amino acids and in DNA/RNA:



histidine



tryptophan



proline

Histidine:

Essential part of many enzyme active sites as nucleophilic catalyst and for H-transfer and ligand in metalloenzymes

Tryptophan:

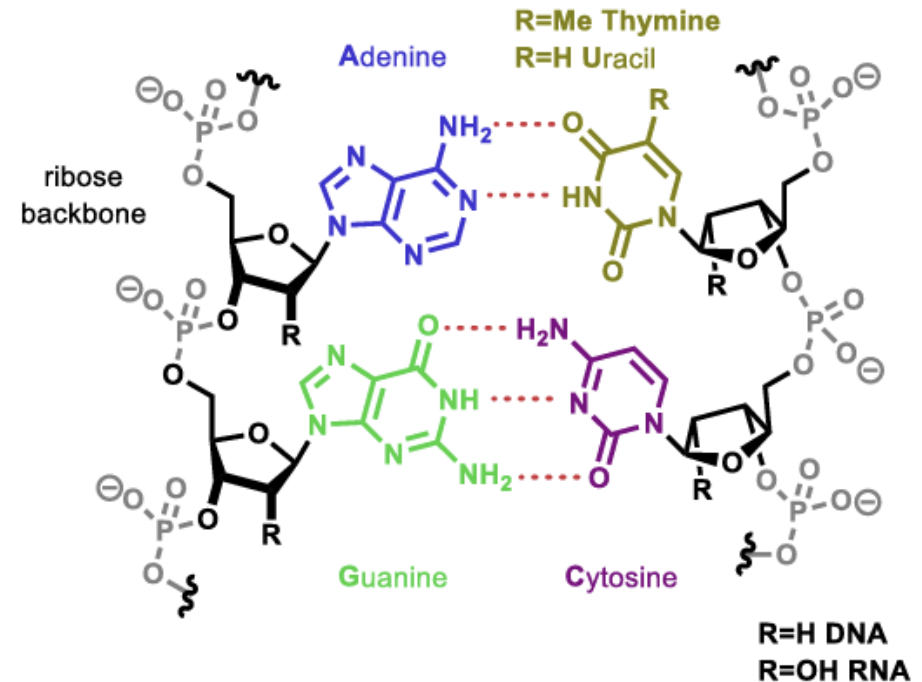
Precursor to neurochemicals

Precursor to vitamin B3

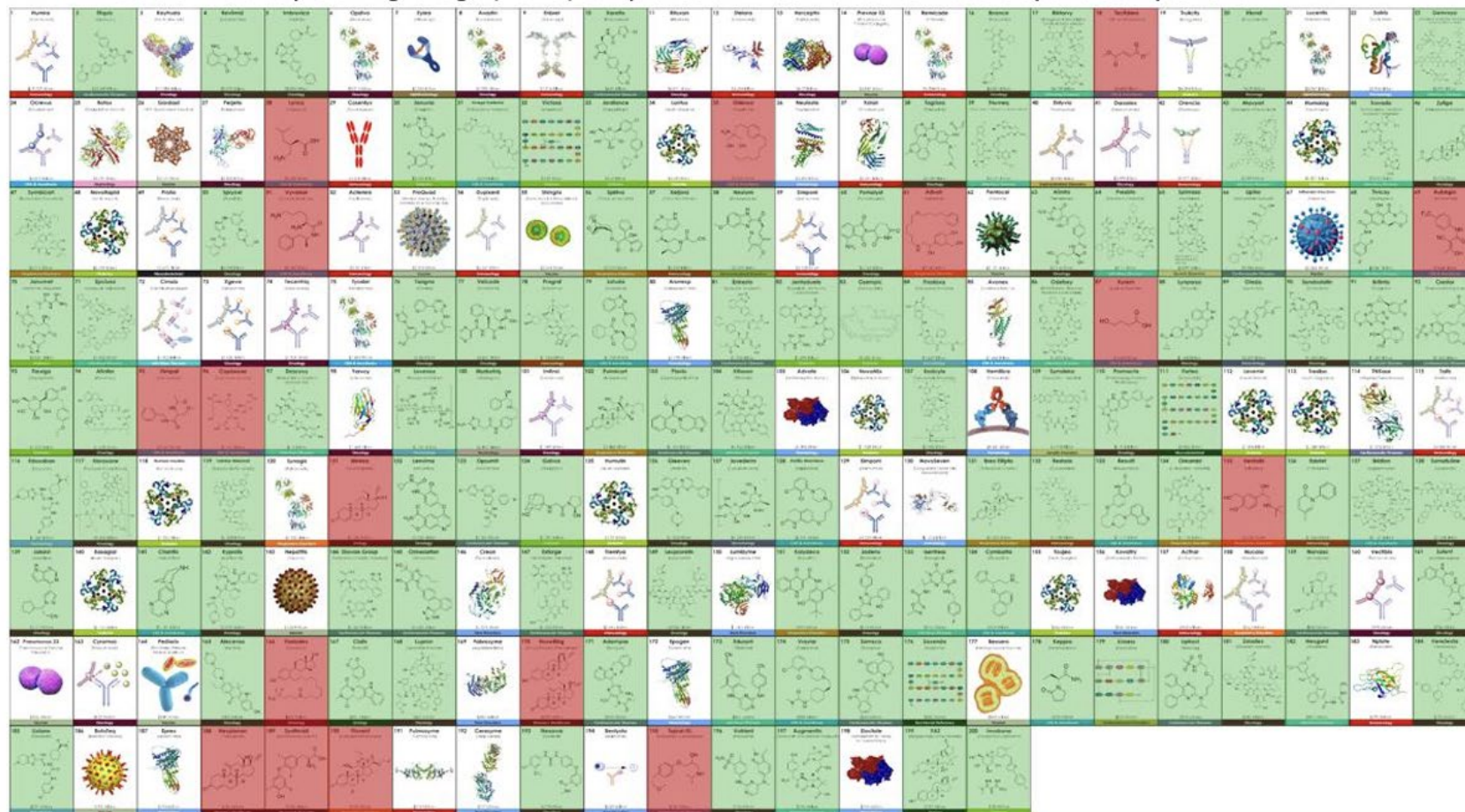
Proline:

Only secondary amine amino acid and brings added rigidity or disruption to protein structure

DNA and RNA:



- **Drug Molecules:**
Out of 200 top selling drugs (2019) only those in red have no heterocyclic component



Green

Drugs that contain a heterocycle

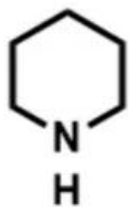
White

Biologics – proteins and antibodies that contain heterocycles as e.g. amino acids

Red

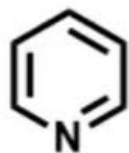
Drugs that contain no heterocycle

Nitrogen-Containing Heterocycles in Drug Molecules



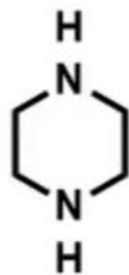
piperidine

11%



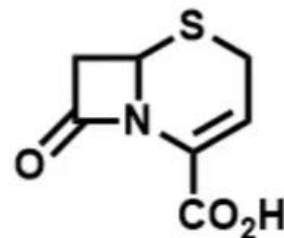
pyridine

10%



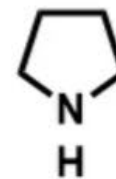
piperazine

9%



CEPHEM

6%



pyrrolidine

6%

From 1086 FDA-approved unique small molecule drugs, 640 contain N-heterocycles

J. Med. Chem. **2014**, 57, 10257

Summary

Definition of a Heterocycle

*cyclic compounds having as ring members
atoms of at least two different elements*

•

Heterocycles in Life and Society

- Polymers and Materials
- Everyday Items
- Amino Acids and DNA / RNA
- Natural Products
- Drug Molecules

IUPAC Rules

Three rule systems for naming heterocycles

I. Retained Names

Older heterocycles keep their traditional names

II. Hantzsch-Widman Nomenclature

Logical system of nomenclature. It is a type of systematic chemical nomenclature used for naming heterocyclic parent hydrides having *no more than ten ring atoms*.

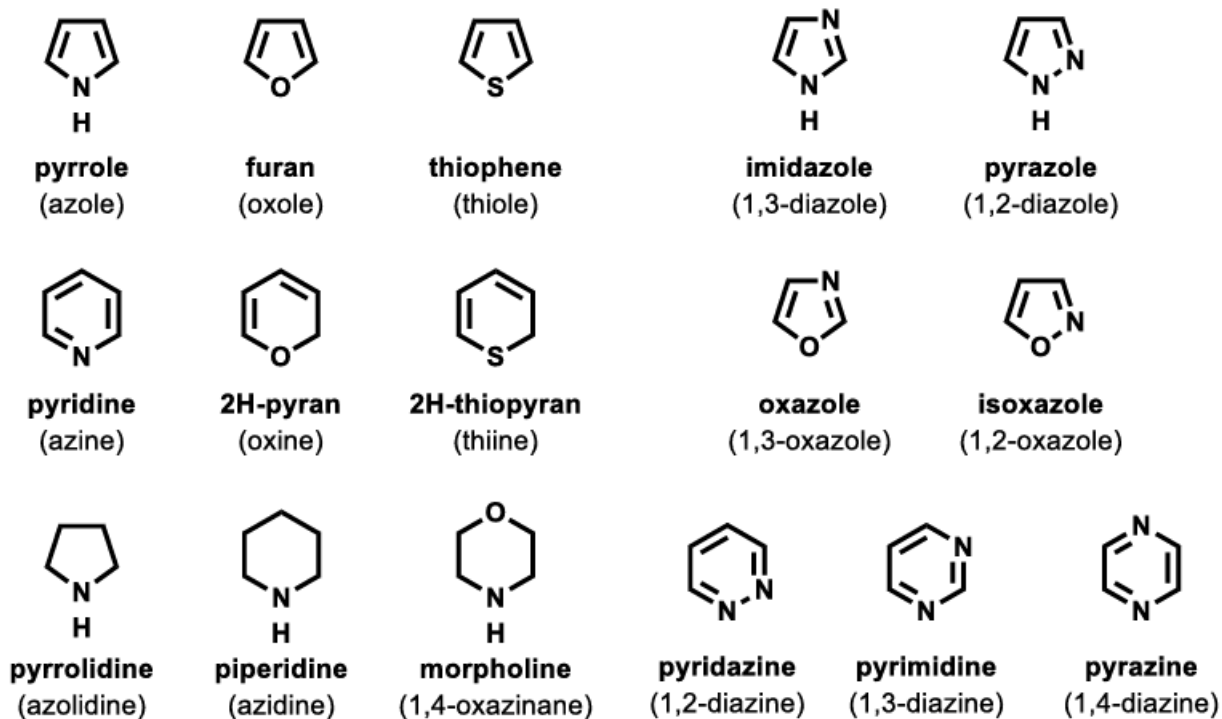
III. Carbon-Framework Substitution Nomenclature (Replacement Nomenclature)

Complex heterocycles

I. Retained Names

Many heterocycles, especially amines, were identified early on, and received trivial names ***which are still preferred***.

Several important heterocycles predate the Hantzsch-Widman nomenclature and keep their **historic names, including:**



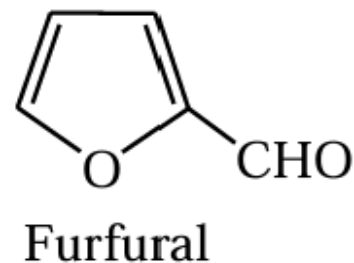
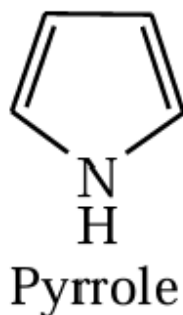
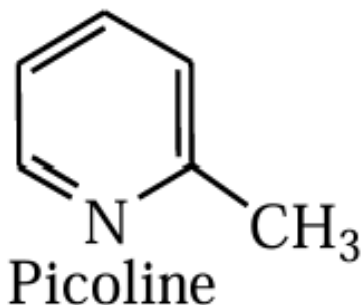
The trivial name is in bold, and a systematic name based on the Hantzsch-Widman system is given beneath it is not bold.

I. Retained Names

Many heterocycles, especially amines, were identified early on, and received trivial names ***which are still preferred***.

Names of the heterocyclic organic compounds were given based on their occurrence, their first preparation and some characteristic properties.

- For example picoline; picoline is derived from coal-tar. This is based on Latin word *pictus* means tarry.
- Pyrrole; which is basic in nature; the name of pyrrole was originated from the Greek word for fiery red because of characteristic colour which the compound gives with pine splint dipped in hydrochloric acid.
- Similarly, the name furfural is given based on its source. Furfural means barn oil. Furfural was isolated from the distillation of barn.



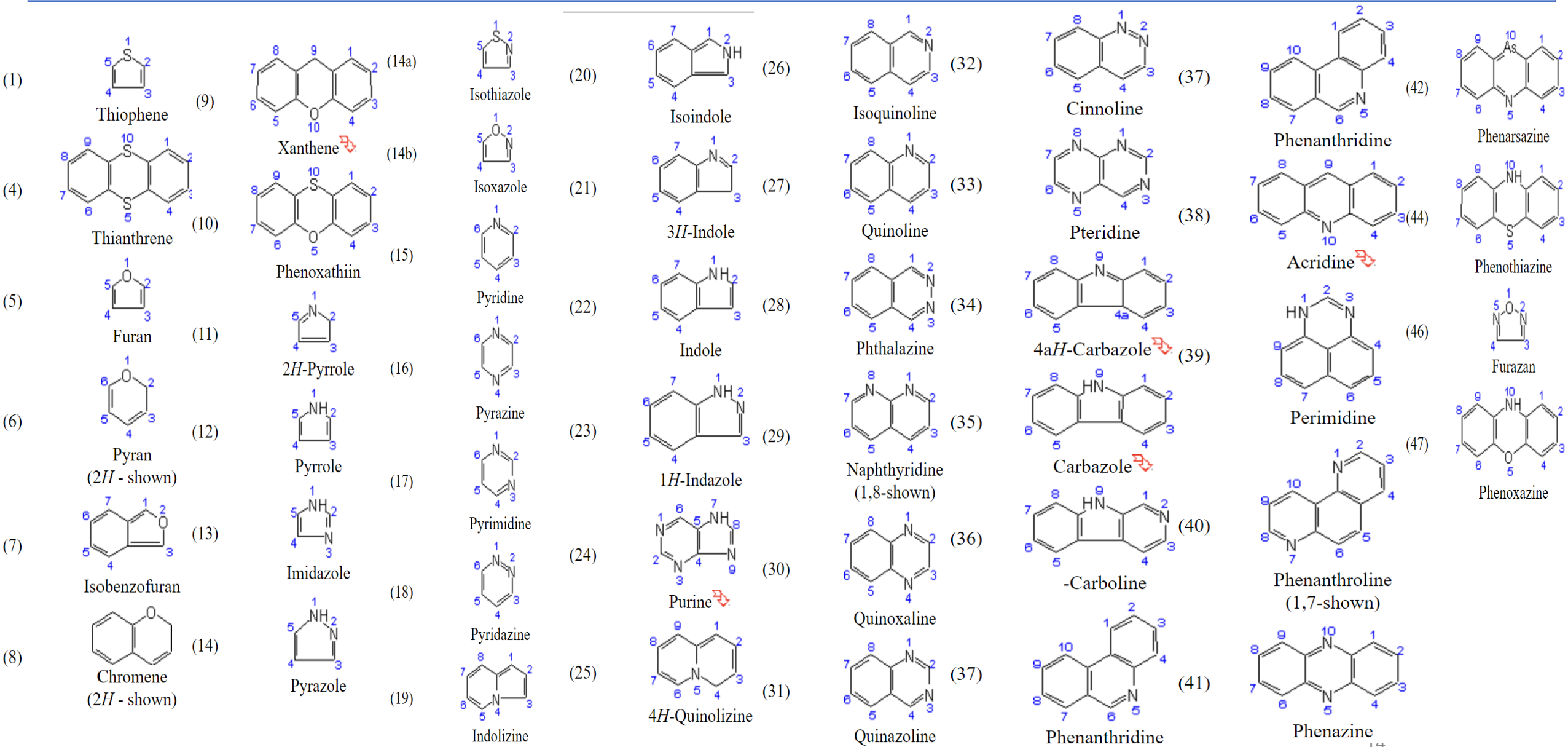
I. Retained Names

At present, just over **60 trivial names recognized by the IUPAC system** of nomenclature.

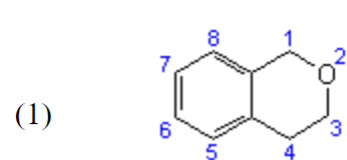
These recognized names are used as a basis for constructing the names of other compounds, systematic names for polycyclic compounds and/or their derivatives.

The next two slides (№ 14, 15) contain an exhaustive list of these structures and their trivial names

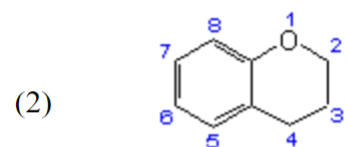
The following trivial and semi-trivial names constitute a partial list of names, which are retained for forming fusion names.



The following trivial and semi-trivial names are retained but are not recommended for use in fusion names.



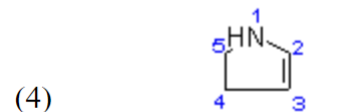
Isochroman



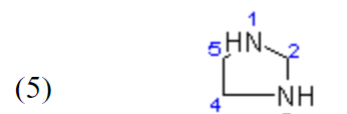
Chroman



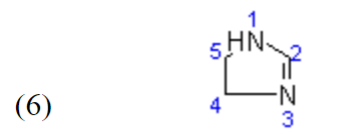
Pyrrolidine



Pyrroline
(2 - shown)



Imidazolidine



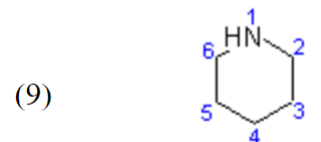
Imidazoline
(2 - shown)



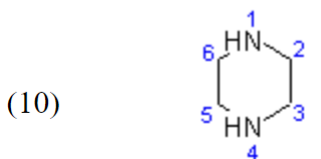
Pyrazolidine



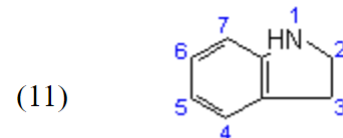
Pyrazoline
(3 - shown)



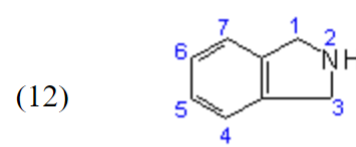
Piperidine



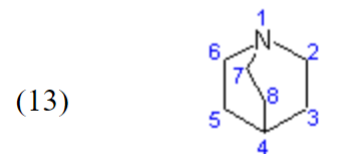
Piperazine



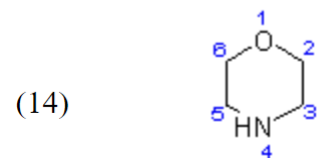
Indoline



Isoindoline



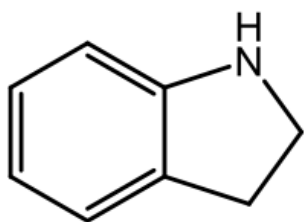
Quinuclidine



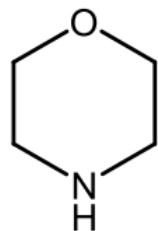
Morpholine

Task 1.

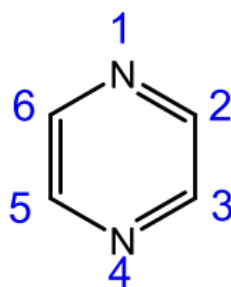
These compounds and their trivial names were presented in Prof. Müller's lecture. Please go through the code and try to match the names and structures. 5 minutes



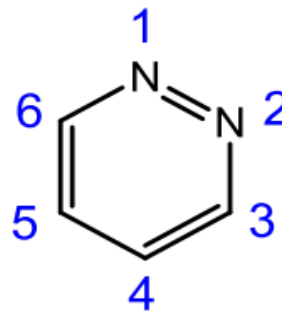
A



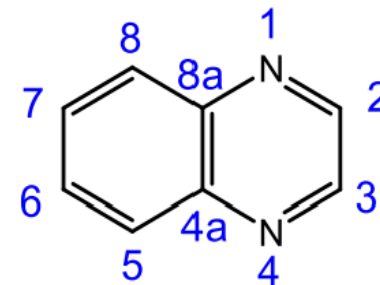
B



C



D



E

<https://forms.gle/SoVJ7uYDt7budwt26>



II. Hantzsch-Widman Nomenclature (system)

This is the most widely used nomenclature system for **monocyclic heterocyclic compounds**, especially for **three to ten-membered ring systems**. The systematic nomenclature gives important structural information.

There is a carefully formulated system for the nomenclature of heterocycles based on **heteroatom(s)**, **ring size**, and **degree of saturation**.

- 1. Heteroatom Element(s)**
Start of name
- 2. Ring Size**
Middle of name
- 3. Amount of Saturation**
End of name

Hantzsch-Widman rules for fully saturated and fully unsaturated heterocycles

I. Type of the heteroatom

The type of heteroatom is indicated by a prefix as shown below for common heteroatoms:

Heteroatom	Prefix
O	oxa-
S	thia-
N	aza- (often imid-, pyr-, pyrid-)
P	phospha-
Si	sila-

O>S>N>P>Si (priority in case of heteroatoms)

II. Ring size (n)

The ring size is indicated by a suffix according to Table below. Some of the syllables are derived from Latin numerals, namely **ir** from *tri*, **et** from *tetra*, **ep** from *hepta*, **oc** from *octa*, **on** from *nona*, **ec** from *deca*.

Ring Size	Suffix	Ring Size	Suffix
3	-ir-	7	-ep-
4	-et-	8	-oc-
5	-ol-	9	-on-
6	-in-	10	-ec-

III. Amount of saturation

The endings indicate the size and degree of unsaturation of the ring.

Ring size	With N		Without N	
	Unsat.	Sat.	Unsat.	Sat.
3	irine	iridine	irene	irane
4	ete	etidine	ete	etane
5	ole	olidine	ole	olane
6	ine	a	in	inane
7	epine	a	epin	epane
8	ocine	a	ocin	ocane
9	onine	a	onin	onane
10	ecine	a	ecin	ecane

a : means use the prefix **perhydro** followed by the fully unsaturated name

IUPAC name = locants + prefix + suffix

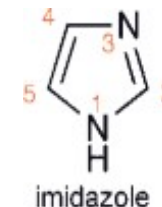
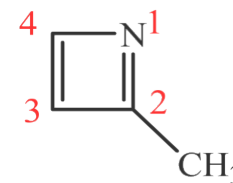
Hantzsch-Widman rules for fully saturated and fully unsaturated heterocycles

1) Identify the heteroatom present in the ring and choose the corresponding **prefix** (e.g. **thia** for **sulfur**, **aza** for **nitrogen** and **oxa** for **oxygen**).

2) The position of a single heteroatom controls the numbering in a monocyclic compound.

The heteroatom is always assigned position **1** and if substituents present are then counted around the ring in a manner so as to take the lowest possible numbers.

3) A **multiplicative prefix** (di, tri, etc.) and locants are used when two or more similar heteroatoms are contained in the ring (two nitrogens indicated by diaza) and the numbering preferably commenced at a saturated rather than an unsaturated atom, as depicted in the following example: **1,3-diazole**:



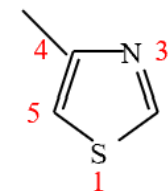
4) If more than one type of heteroatom is present in the ring, the name will include more than one prefix with locants to indicate the relative position of the heteroatoms.

Atom prefixes have a strict order of priority (preference) in which they are to be listed. For example, "**Oxa**" (for oxygen) always comes before "**aza**" (for nitrogen) in a name (see slide 17). When combining the prefixes (e.g. oxa and aza), two vowels may end together; therefore, the vowel at the end of the first part should be omitted (**oxaza**).

The numbering is started from the heteroatom of the highest priority in such a way as to give the smallest possible numbers to the other heteroatoms in the ring (the substituents are irrelevant).

For example:

4-Methyl-**1,3-Thiazaole**



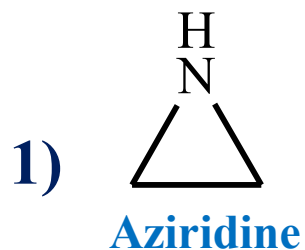
5) Choose the appropriate **suffix** from (table 2) depending on the ring size.

6) The endings indicate the size and degree of unsaturation of the ring.

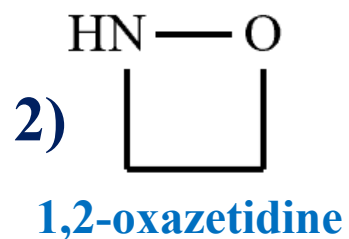
7) Combine the prefix (s) and suffix together and drop the first vowel if two vowels came together.

Hantzsch-Widman rules for fully saturated and fully unsaturated heterocycles

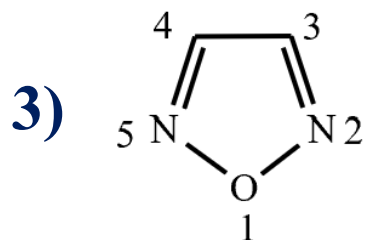
Examples:



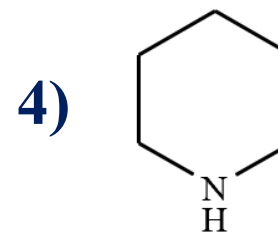
- This ring contains (N): Prefix is **aza**
- The ring is 3-membered and fully saturated: suffix is **iridine**
- By combining the prefix and suffix, two vowels ended up together (**azairidine**), therefore the vowel on the end of the first part should be dropped. This gives the correct name:
Aziridine



- This ring contains (O and N) (O has higher priority than (N) and by starting numbering the ring at (O): Prefix is **1,2-Oxaaza**, but the first vowel must be omitted to give: **1,2-Oxaza**
- The ring is 4-membered and fully saturated: suffix is **etidine**
- By combining the prefix and suffix, two vowels ended up together (**1,2-oaxazaetidine**), therefore the vowel on the end of the first part should be dropped. This gives the correct name:
1,2-oxazetidine



Oxa + diaza + ole = **1,2,5-Oxadiazole**



Perhydro + aza + ine = **perhydroazine**

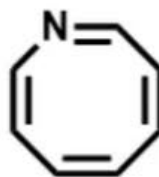
Heterocycle Nomenclature



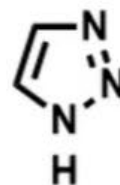
aziridine



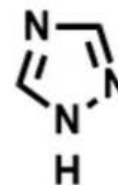
oxepane



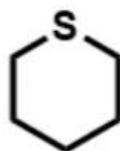
azocene



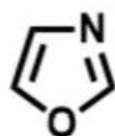
1,2,3-triazole



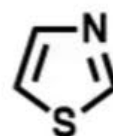
1,2,4-triazole



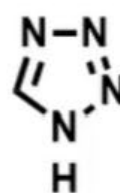
thiane



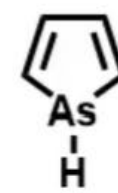
oxazole



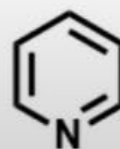
thiazole



tetrazole



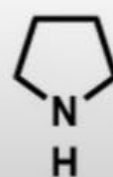
arsole!



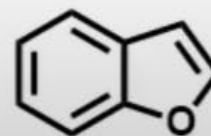
azinane



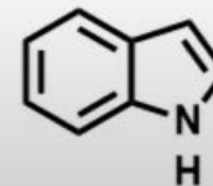
diazole



azolidine



???

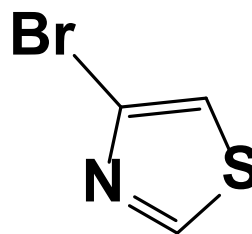
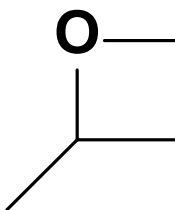
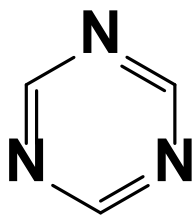
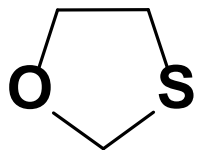


????

Task 2.

Exercise:

Name the following compounds.



A

B

C

D

E



<https://forms.gle/f6RUCcnU9nCfjUfj7>

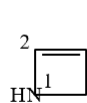
Hantzsch-Widman rules for partially unsaturated heterocycles

- Partial unsaturation in heterocyclic compounds can be indicated by one of the following methods:

- The position of nitrogen or carbon atoms which bear extra hydrogen atoms must be indicated by numbers and italic capital H (e.g. **1*H***, **2*H***, etc.) followed by the name of maximally unsaturated ring.
- The words dihydro, or trihydro, or tetrahydro are used if two or three or four atoms are saturated. These words are preceded by numbers indicate the position of saturated atoms as low as possible and followed by the corresponding fully unsaturated Hantzsch-Widman name.
- Alternatively, the partially unsaturated 4 and 5 rings (i.e. rings contain one double bond) are given special Hantzsch-Widman suffixes as in table 3 and the double bond is specified as Δ^1 , Δ^2 , Δ^3 , etc.. Which indicates 1 and 2 ; 2 and 3; 3 and 4 atoms respectively have a double bond.

Name : Δ^x + Prefix + special suffix (x= locant of the double bond)

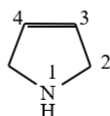
Ring size	With N	Without N
4	etine	etene
5	oline	olene



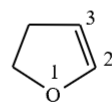
Δ^2 -Azetine
2-Azetine



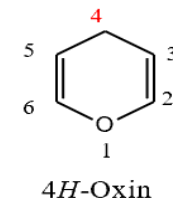
Δ^2 -Oxetene
2-Oxetene



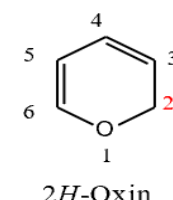
Δ^3 -Azoline
3-Azoline



Δ^2 -Oxolene
2-Oxolene

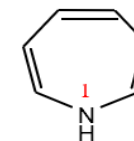


4*H*-Oxin

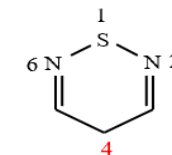


2*H*-Oxin

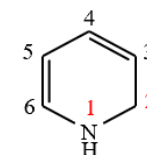
Not 6*H*-Oxin



1*H*-Azepine

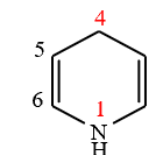


4*H*-1,2,6-Thiadiazine

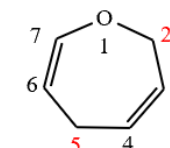


1,2-Dihydroazine

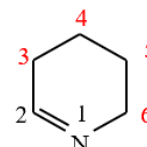
Not 1,6-Dihydroazine



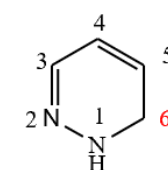
1,4-Dihydroazine



2,5-Dihydrooxepin

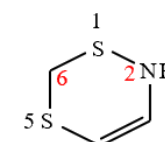


3,4,5,6-Tetrahydroazine
Not 2,3,4,5-Tetrahydroazine



1,6-Dihydro-1,2-diazine

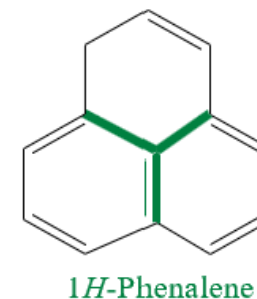
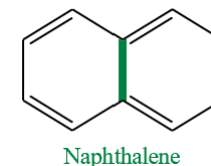
Hint:



2*H*,6*H*-1,5,2-Dithiazine

Definitions:

- **Fusion:** This term is used to describe the process of joining two separate rings with the maximum number of non-cumulative double bonds *via* two atoms and one common bond.
- **Ortho-fused rings:** are those rings that have only two common atoms and one bond, example; **Naphthalene**
- **Ortho-and peri-fused rings:** are those found in a polycyclic compound with a ring that is *ortho*-fused to different sides of two other rings that are themselves *ortho*-fused together (i.e. there are three common atoms between the first ring and the other two), example; **1H-phenalene** is considered as being composed of three benzene rings, each is *ortho-peri*-fused to the other two.
- **Polycyclic compounds** incorporating one heterocyclic ring or fused heterocyclic system fused are known **benzoheterocycles**.
- Also bicyclic compounds with two fused heterocyclic rings are well known.
- Both types can be named according to certain rules.



Nomenclature of Fused Systems

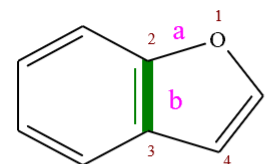
A. Nomenclature of benzofused compounds:

- Unless listed as a trivially named heterobicycle (see slide 6), a benzene ring fused to a heteromonocycle of five or more members or a heterobicyclic system is named by prefixing the word **benzo** to a letter indicating the position of fusion in square brackets by the name of heterocyclic ring (common or IUPAC or modified replacement name).

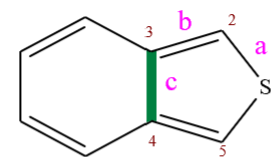
Name = Benzo [letter] name of heterocyclic ring

- For designating the position of fusion, the peripheral bonds of the heterocyclic ring are consecutively assigned alphabetical letters starting with the 1,2-bond as a side and the labeling is continued around the ring to give the common bond the lowest order.

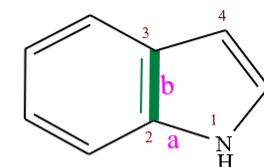
Examples:



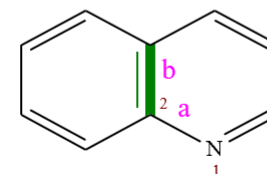
Benzo[b]furan



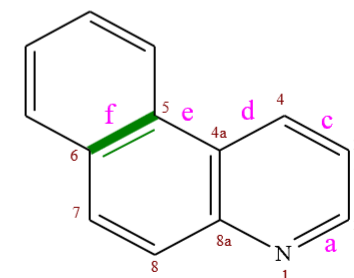
Benzo[c]thiophene



Benzo[b]pyrrole
Indole



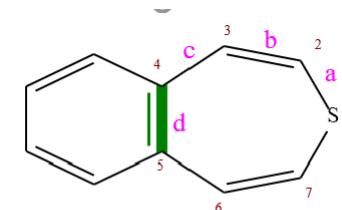
Benzo[b]pyridine
Quinoline



Benzo[f]quinoline

Nomenclature of Fused Systems

- There is An exception to the two ring systems in which a benzene ring is fused to a hetero ring (which doesn't have a known common name) may be named by prefixing numbers indicating the positions of the hetero atoms to benzo followed by the name of the heterocyclic component.
- Numbering is assigned according to priority order of the hetero atoms i.e.



Benzo[d]thiophene

B. Nomenclature of fused heterocyclic compounds:

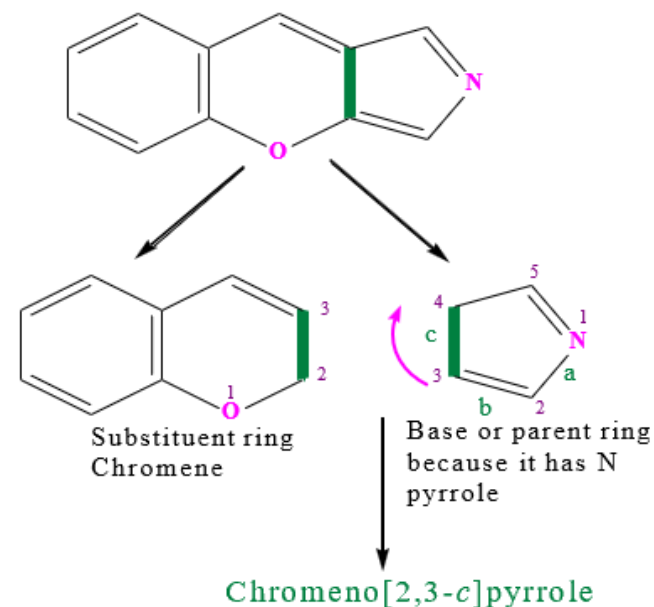
- Naming a fused heterocyclic systems composed of two monoheterocyclic units or benzoheterocycles (e.g. chromene) fused with another heterocycle ring is based upon considering one system as the parent (base) and the second is considered as substituent.
- Name : name of minor ring [number, number-letter] name of major ring**
- The name of the minor ring is derived by writing a contracted prefix for the substituent ring present. In an attached component prefix the terminal 'e' is changed to 'o' with exception of:

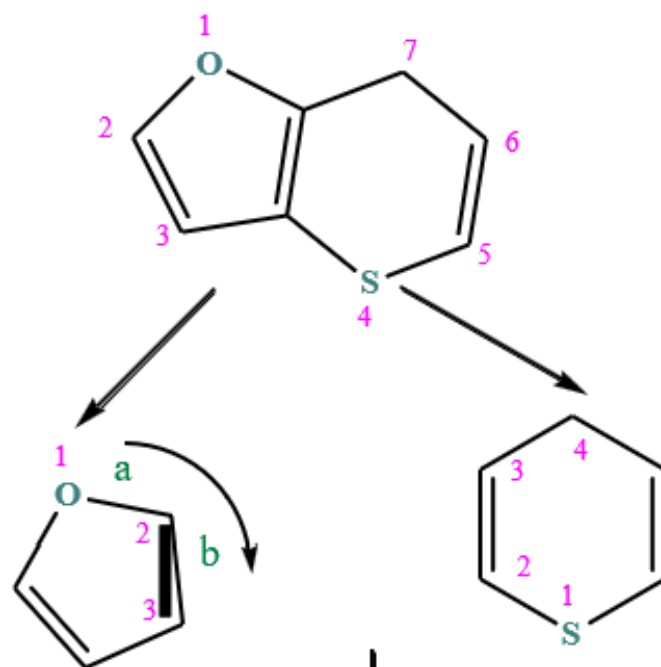
Furo	From Furan
Imidazo	From Imidazole
Thieno	From Thiophene
pyrido	From Pyridine
Pyrimido	From Pyrimidine
Quino	From Quinoline
Isoquino	From Isoquinoline

Nomenclature of Fused Systems

- The **numbers** indicate which atoms in the minor ring are common to the major ring (fusion sites in minor ring).
- The order of the numbers indicates which atom of the minor ring is encountered closest to atom **1** in the major numbering system (i.e. these numbers may be written in ascending or descending order e.g. 2,3 or 3,2)
- The **letter** defines the position of attachment of the minor ring to the major ring (fusion sites in base component).
- Finally a suffix indicate the name of the base ring is written.
- The numbering system for the whole fused system is not the same as the numbers in the square brackets (i.e. there are three numbering systems; one for minor ring, one for major ring and the third is for the system as a whole).
- **Priority order of component ring systems:** Selection of a parent component or attached component is based on the following rules which are applied in order:

Rule 1: A heterocyclic ring containing the heteroatom occurring earliest in the order **N**, F, Cl, Br, I, **O**, **S**, Se,.. (i.e. ring containing N preferred to the rings does not contain N or containing O, or S)





Parent ring

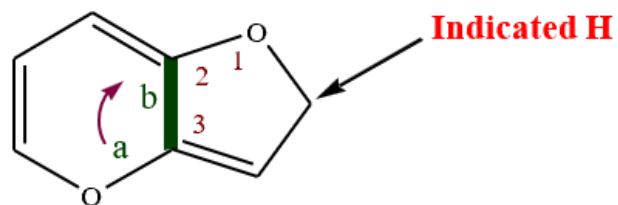
O preferred to S
Furan

Substituent ring

Thiopyrano with one saturated C that take locant 7 when the system is numbered as a whole (starting from O to give the two heteroatoms locants 1,4 while starting from S gives them locants 1,5)

7H-Thiopyrano[3,2-*b*]furan

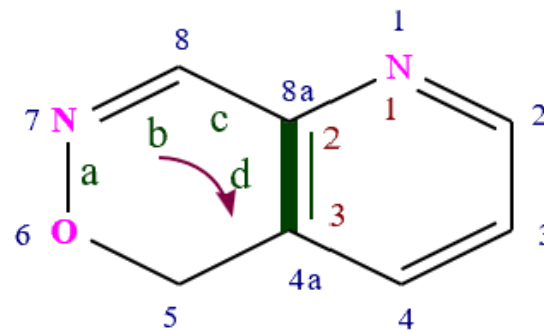
Rule 2: A heterocyclic component containing the largest possible individual ring



*2H-Furo[3,2-*b*]pyran*
(pyran [6] preferred to furan [5])

Numbering the whole system is started from O in furan ring to give the two heteroatoms locants 1,4 while starting from O in pyran ring gives them locants 1,5, thus the indicated H takes locant 2

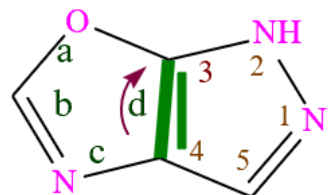
Rule 3: A heterocyclic component containing the greater number of heteroatoms of any kind



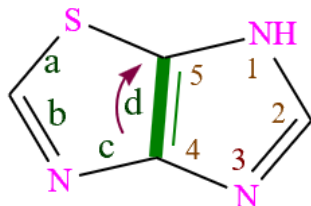
*5H-Pyrido[2,3-*d*][1,2]oxazine*
(Oxazine preferred to pyridine)

N.B. The whole molecule is numbered starting from pyridine ring to give the three heteroatoms the lowest locants (1,6,7), however, stating from oxazine ring will give them locants (2,3,5) or (2,3,8).

Rule 4: A heterocyclic component containing the greater variety of hetroatoms



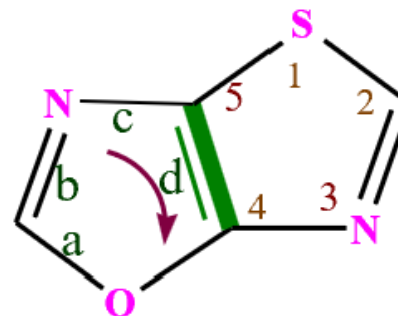
1H-Pyrazolo[4,3-*d*][1,3]oxazole
(O & N preferred to N only)



1H-Imidazo[4,5-*d*][1,3]thiazole

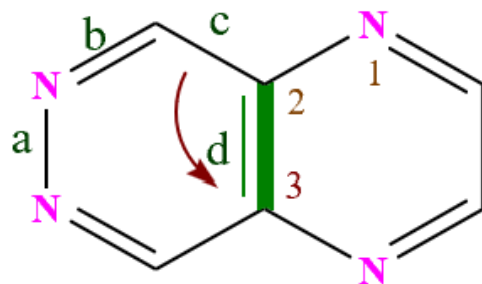
N.B. The whole molecule is numbered starting from pyrazole ring to give the four heteratoms the lowest locants (1,2,4,6). While starting from oxazole ring give them locants (1,3,4,5) or (1,3,5,6).

Rule 5: A heterocyclic component containing the greater number of heteroatoms most preferred when considered in order F, Cl, Br, I, O, S, Se, Te, N, P, As, Sb, Bi, Si, Ge, Sn Pb, B, Hg



[1,3]Thiazolo[5,4-*d*][1,3]oxazole
(N & O preferred to N & S)

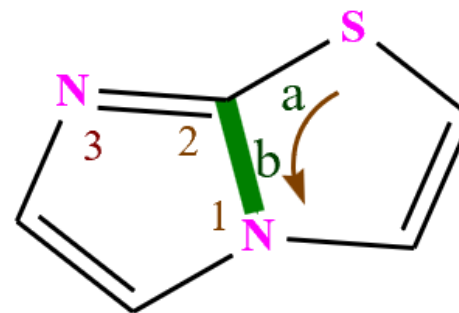
Rule 6: A heterocyclic component with the lower locants for heteroatoms



Pyrazino[2,3-*d*]pyridazine

(pyridazine [2N-1,2] preferred to pyrazine [2N-1,4])

Rule 7: If a position of fusion is occupied by a heteroatom the name of the component rings to be used are so chosen as both to contain the heteroatom.

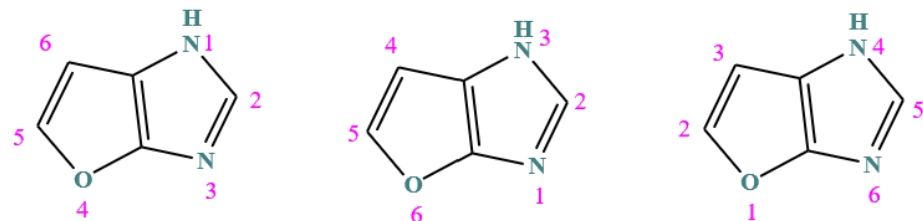


Imidazo[2,1-*b*][1,3]thiazole

Order of preference between alternative numbering system of the whole molecule

- Numbering the whole fused system should start from the first atom after fusion in any direction to fulfill the following rules in order:

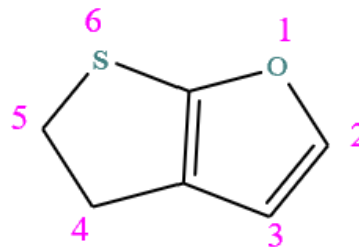
a) Give low numbers for the heteroatoms as a set



1*H*-Furo[2,3-*d*]imidazole

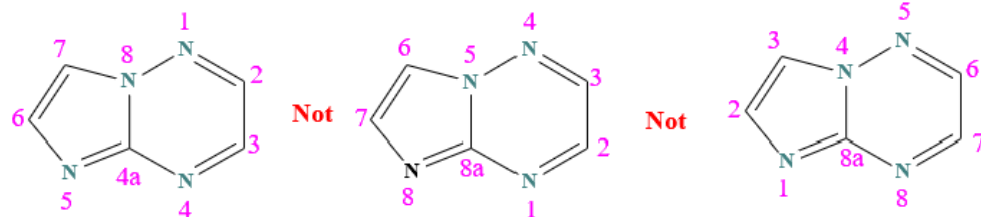
(heteroatoms 1,3,4 is preferred to 1,3,6 or 1,4,6)

b) Give low numbers for heteroatoms of higher priority i.e. O, S, N



4,5-Dihydro-thieno[2,3-*b*]furan

c) Give low numbers to fusion carbon atoms



Imidazo[1,2-*b*][1,2,4]triazine

fusion C -4a is preferred to 8a

d) Give low numbers to indicated hydrogen atom



2*H*,4*H*-[1,3]dioxol[4,5-*d*]imidazole

Indicated hydrogens 4 not 6

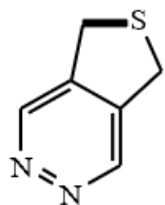
Summary of Nomenclatures Rules

Scheme for deriving the base component of a fused ring system

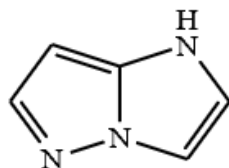
- Is there only one ring which contains nitrogen?
Yes: choose this as base component
- Are the two rings have the same heteroatoms but their size is different ?
Yes: choose the larger one
- Are the two rings of the same size but have different heteroatoms?
Yes: choose the ring containing a heteroatom of the highest priority i.e. $O > S$
- Are the rings of the same size but contain different numbers of heteroatoms?
Yes: choose the ring with the greater number
- Are the two rings of the same size and the same number of different heteroatoms?
Yes: choose the ring with the greatest variety of heteroatoms
- Are the two rings have the same size and the same number and type of heteroatoms?
Yes: choose the ring with the lower numbers for heteroatoms

Task 3.

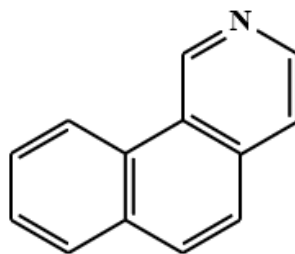
Exercise : Name the following compounds.



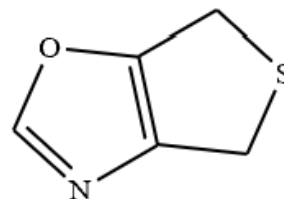
(a)



(b)



(c)

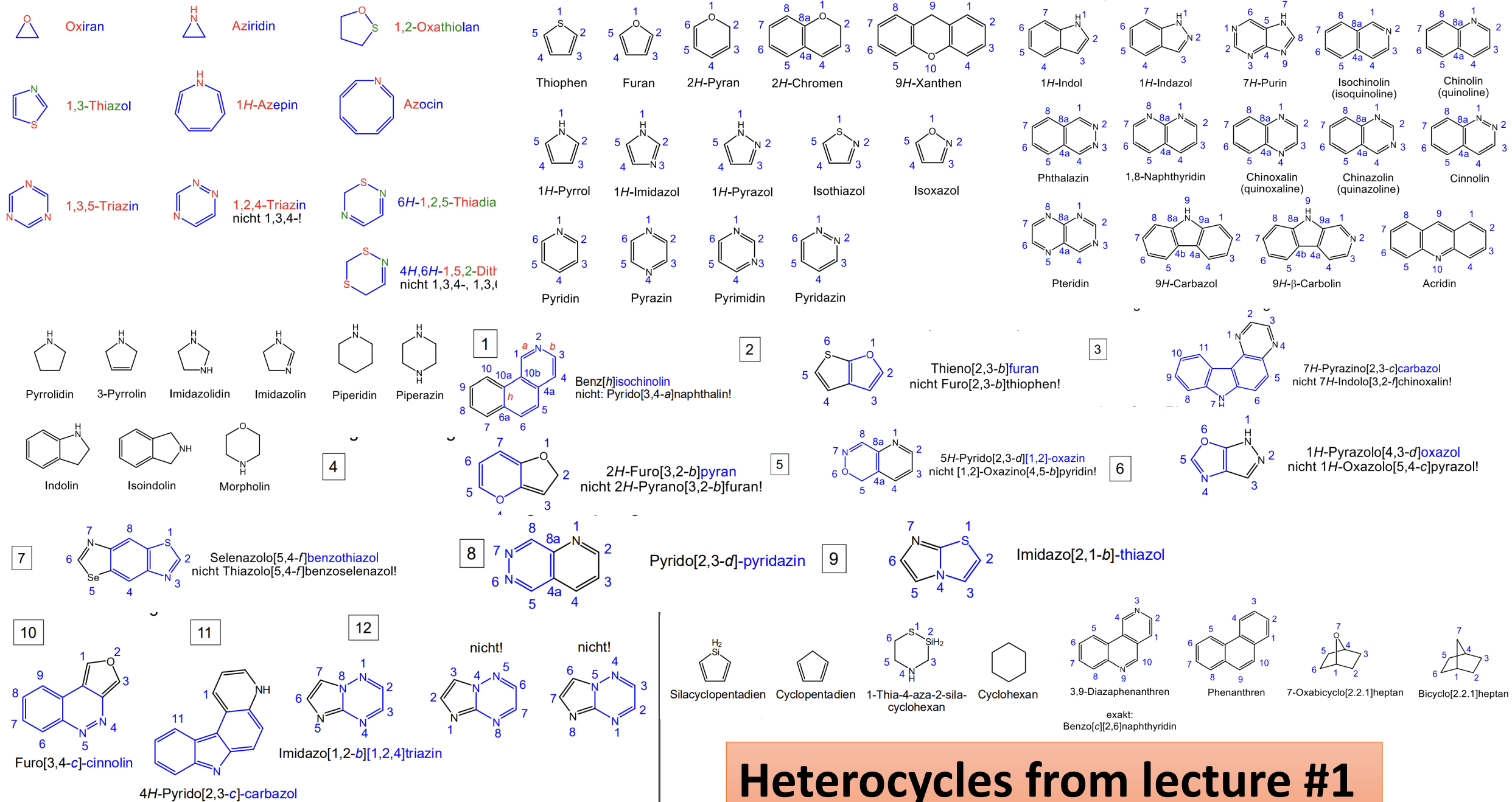


(d)

<https://forms.gle/sJ29iYo46ZV9ZthJ7>



**Thank you for your
attention!**

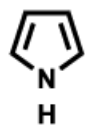


Heterocycles from lecture #1

Additional materials

Retained Names

Several important heterocycles predate the Hantzsch-Widman Nomenclature and these keep their **historic names**, including:



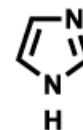
pyrrole
(azole)



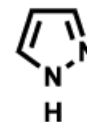
furan
(oxole)



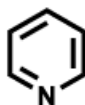
thiophene
(thiole)



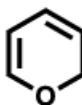
imidazole
(1,3-diazole)



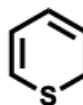
pyrazole
(1,2-diazole)



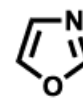
pyridine
(azine)



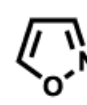
2H-pyran
(oxine)



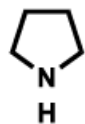
2H-thiopyran
(thiine)



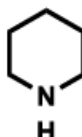
oxazole
(1,3-oxazole)



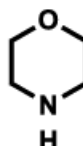
isoxazole
(1,2-oxazole)



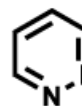
pyrrolidine
(azolidine)



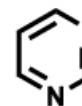
piperidine
(azidine)



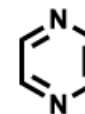
morpholine
(1,4-oxazinanane)



pyridazine
(1,2-diazine)



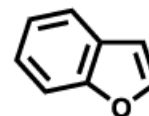
pyrimidine
(1,3-diazine)



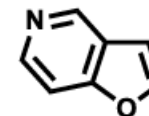
pyrazine
(1,4-diazine)

Fused Heterocycles

If a heterocycle is fused to another system, join the two names together with the biggest and most heteroatom-containing ring as the second part.



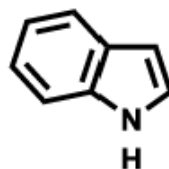
benzofuran
benzo[b]furan
(benzene fused to furan)



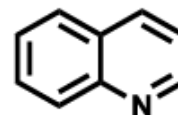
furopyridine
furo[3,2-c]pyridine
(furan fused to pyridine)

Fused Heterocycles with Retained Names

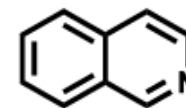
Several important heterocycles predate the Hantzsch-Widman Nomenclature and these keep their **historic names**



indole
(benzopyrrole)



quinoline
(benzo[b]pyridine)

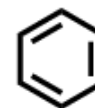


isoquinoline
(benzo[c]pyridine)

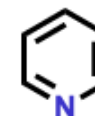
Carbon-Framework Substitution

Used to describe complex (multiple fused together) and large (>10 in ring) heterocycles

1. Name the all carbon version of the heterocycle - there are robust methods for describing complex all-carbon structures
2. Use substitution to describe the positions of the heteroatoms



benzene

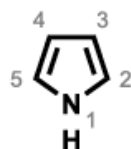


(aza-benzene)
(azine)
pyridine

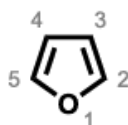
Numbering

Starting at 1 = the HETEROATOM with the:

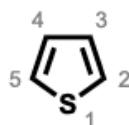
- BIGGEST PERIOD followed by SMALLEST ROW in the periodic table
- If there are multiple heteroatoms, the most saturated one takes the lowest number
- Number in a direction that gives other heteroatoms lowest numbers
- Number in a direction that gives substituents lowest numbers
- Isoquinoline takes the same numbering system as quinoline



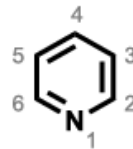
pyrrole



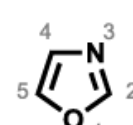
furan



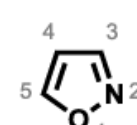
thiophene
(thiole)



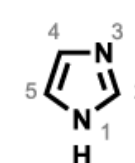
pyridine



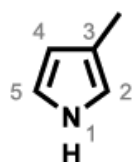
oxazole
(1,3-oxazole)



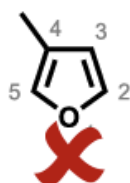
isoxazole
(1,2-oxazole)



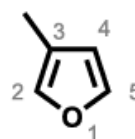
imidazole
(1,2-diazole)



pyrrole

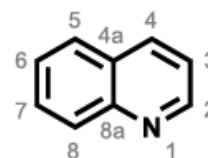


furan

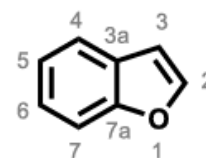


furan

number to give any substituent
the lowest number

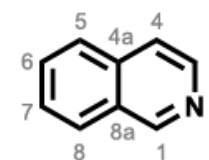


quinoline



benzofuran

numbering **SKIPS**
ring junction carbon atoms

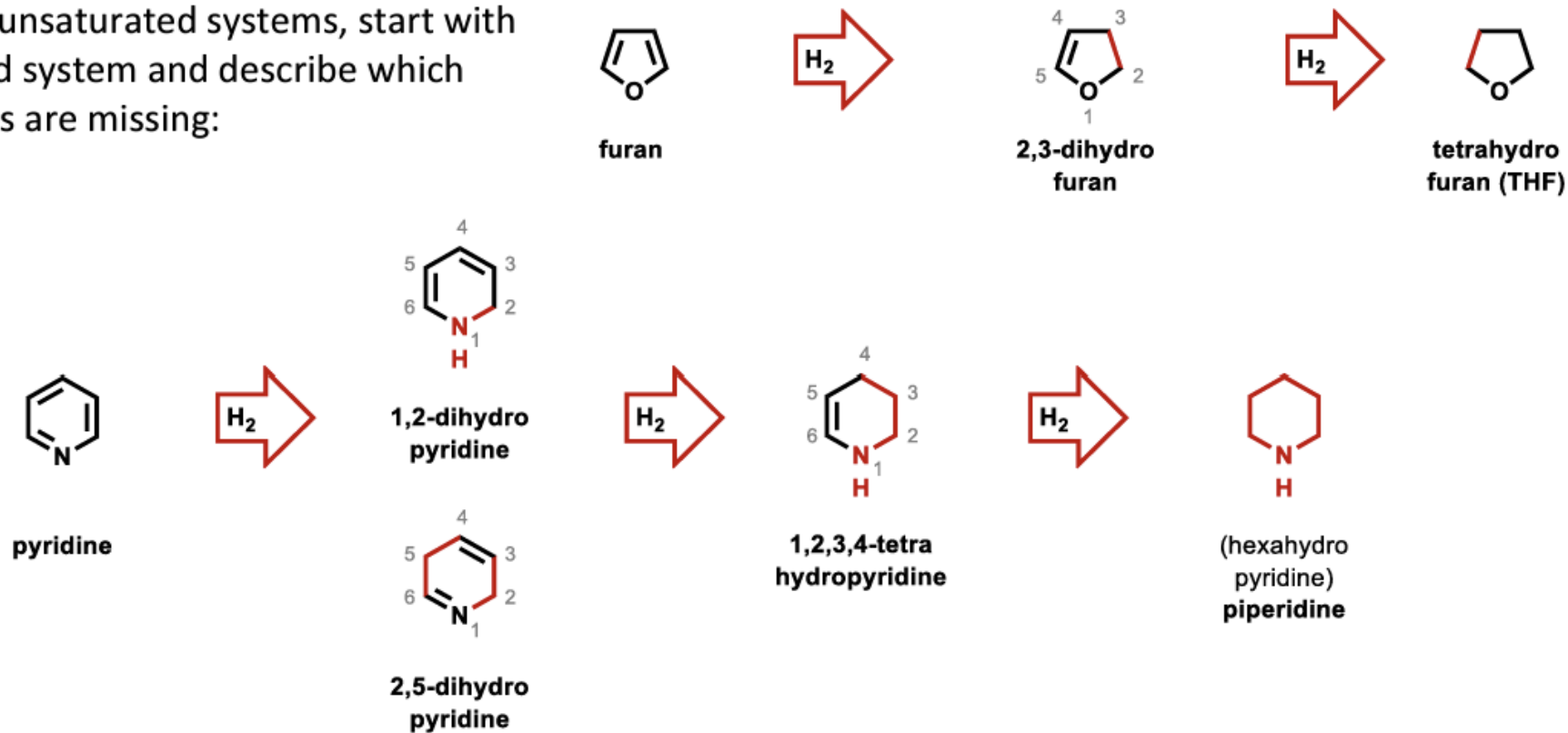


isoquinoline

numbering for isoquinoline
is the **SAME** as quinoline and
does not start with N=1

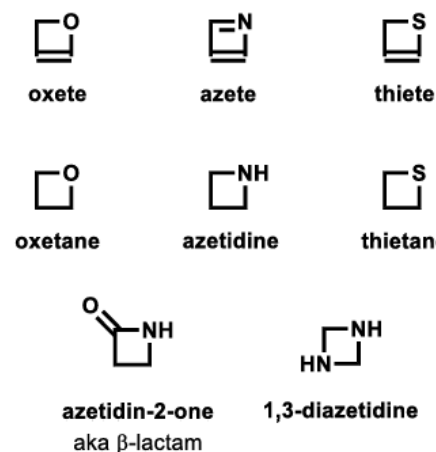
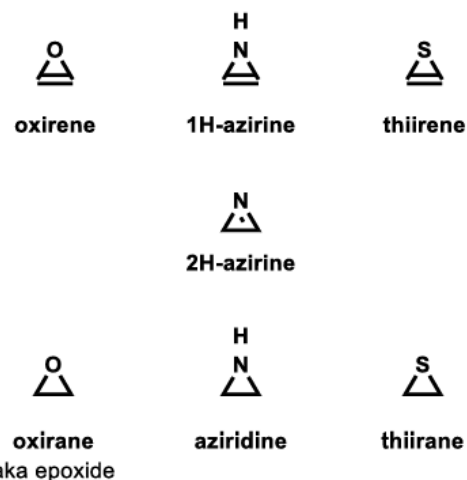
Degree of Unsaturation

For partially unsaturated systems, start with the saturated system and describe which double bonds are missing:

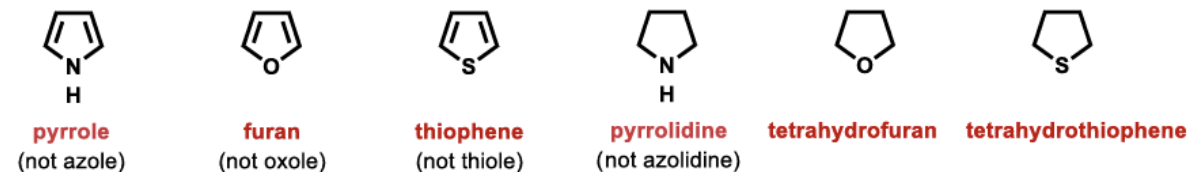


Appendix A: Selected Heterocycles

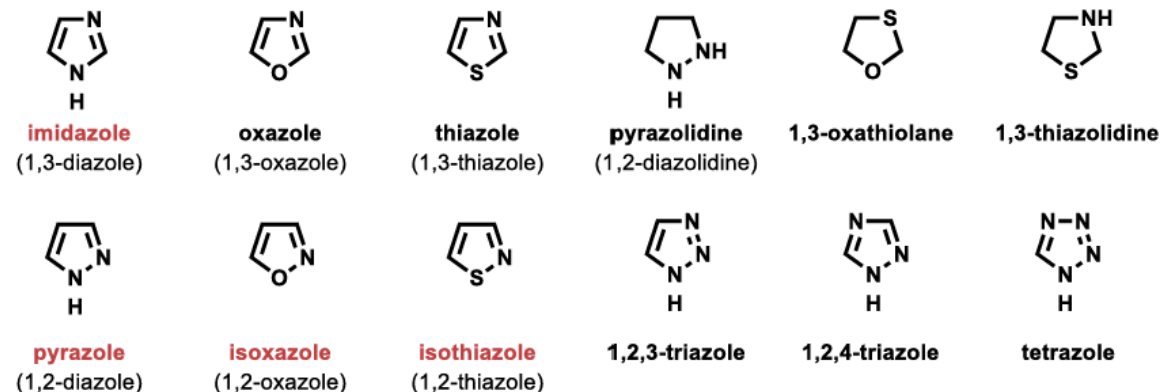
3 IR



4 ET



5 OL



6 IN



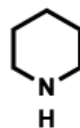
pyridine
(not azine)



2H-pyran
(not oxine)



2H-thiopyran
(thiine)



piperidine
(azinane)



tetrahydropyran
(oxane)



thiane



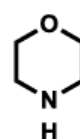
pyridazine
(1,2-diazine)



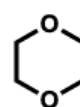
pyrimidine
(1,3-diazine)



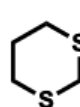
pyrazine
(1,4-diazine)



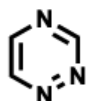
morpholine
(1,4-oxazinane)



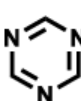
1,4-dioxane



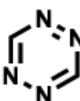
1,3-dithiane



1,2,4-triazine



1,3,5-triazine

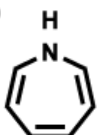


1,2,4,5-tetrazine

7 EP

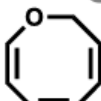


oxepane

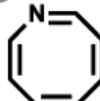


azepine

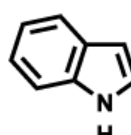
8 OC



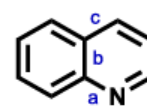
2H-oxocine



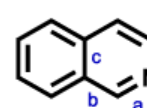
azocine



indole
(benzopyrrole)



quinoline
(benzo[b]pyridine)



isoquinoline
(benzo[c]pyridine)

Retained name in **red**, IUPAC version of retained name in (brackets).

More at:

<http://www.compoundchem.com/wp-content/uploads/2014/07/Heterocycles-graphic.png>

https://en.wikipedia.org/wiki/Heterocyclic_compound

HETEROCYCLES IN ORGANIC CHEMISTRY

A HETEROCYCLE IN ORGANIC CHEMISTRY IS A RING OF CONNECTED ATOMS, WHERE ONE OR MORE OF THE ATOMS IN THE RING ARE ELEMENTS DIFFERENT FROM CARBON. HETEROCYCLES WITH OXYGEN, NITROGEN, AND SULFUR ARE THE MOST PREVALENT; SELENIUM, BORON, SILICON, ARSENIC & PHOSPHORUS CAN ALSO BE INCORPORATED.

 OXYGEN-BASED HETEROCYCLES

 NITROGEN-BASED HETEROCYCLES

 SULFUR-BASED HETEROCYCLES

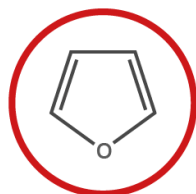
 MULTIPLE HETEROATOM HETEROCYCLES



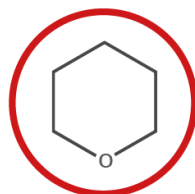
ETHYLENE OXIDE
Oxirane
 C_2H_4O



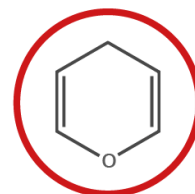
TETRAHYDROFURAN
Oxolane
 C_4H_8O



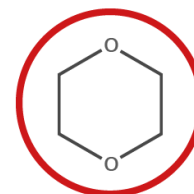
FURAN
Oxole
 C_4H_4O



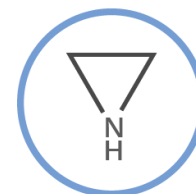
TETRAHYDROPYRAN
Oxane
 $C_5H_{10}O$



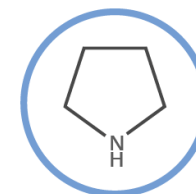
4H-PYRAN
4H-Oxine
 C_5H_6O



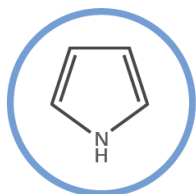
1,4-DIOXANE
p-Dioxane
 $C_4H_8O_2$



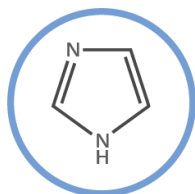
ETHYLENE IMINE
Aziridine
 C_2H_3N



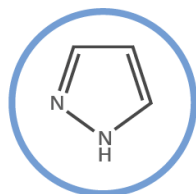
PYRROLIDINE
Azolidine
 C_4H_9N



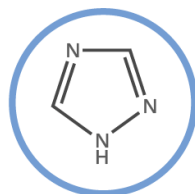
PYRROLE
Azole
 C_4H_5N



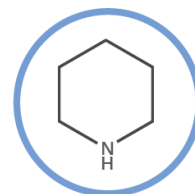
IMIDAZOLE
1,3-diazole
 $C_3H_4N_2$



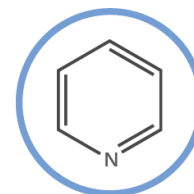
PYRAZOLE
1,2-diazole
 $C_3H_4N_2$



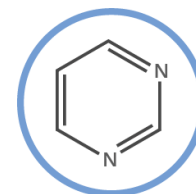
TRIAZOLE
1,2,4-triazole
 $C_2H_3N_3$



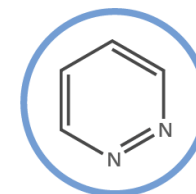
PIPERIDINE
Azinane
 $C_5H_{11}N$



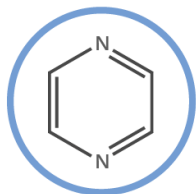
PYRIDINE
Azine
 C_5H_5N



PYRIMIDINE
1,3-diazine
 $C_4H_4N_2$



PYRIDAZINE
1,2-diazine
 $C_4H_4N_2$



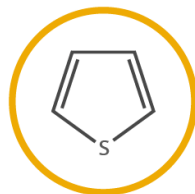
PYRAZINE
1,4-diazine
 $C_4H_4N_2$



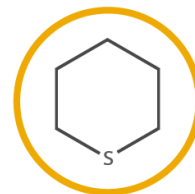
ETHYLENE SULFIDE
Thiirane
 C_2H_4S



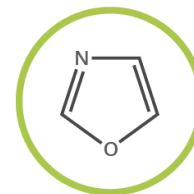
TETRAHYDROTHIOPHENE
Thiolane
 C_4H_8S



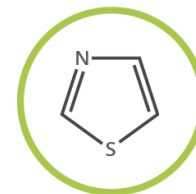
THIOPHENE
Thiole
 C_4H_4S



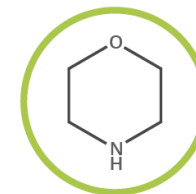
TETRAHYDROTHIOPYRAN
Thiane
 $C_5H_{10}S$



OXAZOLE
1,3-oxazole
 C_3H_3NO



THIAZOLE
1,3-thiazole
 C_3H_3NS



MORPHOLINE
Tetrahydro-1,4-oxazine
 C_4H_9NO